Black-Scholes-Merton Equation and Risk-Neutral Pricing

Yao Xiao | Xinqi Zou | Yingxuan Li

NYU Shanghai
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Brownian Motion

Definition 1.1.

Let X be a random variable defined on a nonempty sample space Ω. Let $\mathcal{G}$ be a σ-algebra of subsets of Ω. If every set in $\sigma(X)$ is also in $\mathcal{G}$, we say that X is $\mathcal{G}$-*measurable*.

Definition 1.2.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, for each $\omega \in \Omega$, suppose there is a continuous function $W(t)$, $t \geq 0$ that depends on ω. Then $W(t)$, $t \geq 0$, is a *Brownian motion* if for all $0 = t_0 < t_1 < \ldots < t_m$ s.t.

(i) $W(0) = 0$;

(ii) $W(t_i) - W(t_j) \sim N(0, t_i - t_j)$; and

(iii) $W(t_i) - W(t_j)$ are independent.

Filtration

Definition 1.3.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which is defined a Brownian motion $W(t)$, $t \geq 0$. A *filtration for Brownian motion* is a collection of σ-algebras $\mathcal{F}(t)$, $t \geq 0$, satisfying:

(i) **(Information accumulates)** For $0 \leq s < t$, every set in $\mathcal{F}(s)$ is also in $\mathcal{F}(t)$;

(ii) **(Adaptivity)** For each $t \geq 0$, the Brownian motion $W(t)$ at time t is $\mathcal{F}(t)$-measurable; and

(iii) **(Independence of future increments)** For $0 \leq t < u$, the increment $W(u) - W(t)$ is independent of $\mathcal{F}(t)$.

Martingale

Definition 1.4.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let T be a fixed positive number, and let $\mathcal{F}(t)$, $0 \leq t \leq T$, be a filtration of sub-σ-algebras of $\mathcal{F}$. Consider an adapted stochastic process $M(t)$, $0 \leq t \leq T$. If $\mathbb{E}(M(t)|\mathcal{F}(s)) = M(s)$ for all $0 \leq s \leq t \leq T$, then we say that this process is a *martingale*.

Martingale property of Brownian motion

Theorem 1.5.

Brownian motion is a martingale.

Proof.

Let $0 \leq s \leq t$ be given. Then

$$\begin{aligned}\mathbb{E}(W(t)|\mathcal{F}(s)) &= \mathbb{E}((W(t) - W(s)) + W(s)|\mathcal{F}(s)) \\ &= \mathbb{E}(W(t) - W(s)|\mathcal{F}(s)) + \mathbb{E}(W(s)|\mathcal{F}(s)) \\ &= \mathbb{E}(W(t) - W(s)) + W(s) = W(s)\end{aligned}$$

□

Quadratic variation

Definition 1.6.

Let $f(t)$ be a function defined for $0 \leq t \leq T$. The *quadratic variation* of f up to time T is

$$[f, f](T) = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} (f(t_{j+1} - f(t_j))^2$$

where $\Pi = \{t_0, t_1, \cdots, t_n\}$ and $0 = t_0 < t_1 < \cdots < t_n = T$.

Quadratic variation

Theorem 1.7.

Let W be a Brownian motion. Then $[W, W](T) = T$ for all $T \geq 0$ almost surely.

Proof (Sketch).

Denote

$$Y_{j+1} = \frac{W(t_{j+1}) - W(t_j)}{\sqrt{t_{j+1} - t_j}}$$

We choose a large number n and take $t_j = jT/n$, $j = 0, 1, \cdots, n$. Then $t_{j+1} - t_j = T/n$ for all j and $(W(t_{j+1}) - W(t_j))^2 = T\frac{Y_{j+1}^2}{n}$

Quadratic variation

Proof (Sketch, Continued).

The law of large number then implies that

$$\sum_{j=0}^{n-1} \frac{Y_{j+1}^2}{n} \to \mathbb{E}(Y_{j+1}^2) \quad \text{as } n \to \infty$$

where $Y_{j+1} \sim N(0,1)$. Since $\mathbb{E}(Y_{j+1}^2) = 1$, then we have

$$\sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2 = \sum_{j=0}^{n-1} T \frac{Y_{j+1}^2}{n} \to T \quad \text{as } \|\Pi\| \to 0$$

This completes the proof. □

Important results

In Theorem 1.7, we see that

$$\lim_{\|\Pi\|\to 0} \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2 = T$$

This means that Brownian motion accumulates quadratic variation at rate one per unit time, and can be written in differential form as $dW(t)dW(t) = dt$. Similar results can be deduced as follows.

Important results

$$
\begin{aligned}
&\lim_{\|\Pi\|\to 0}\left|\sum_{j=0}^{n-1}(W(t_{j+1})-W(t_j))(t_{j+1}-t_j)\right| \\
\leq\ &\lim_{\|\Pi\|\to 0}\sum_{j=0}^{n-1}\max_{0\leq k\leq n-1}|W(t_{k+1})-W(t_k)|(t_{j+1}-t_j) \\
=\ &\lim_{\|\Pi\|\to 0}\max_{0\leq k\leq n-1}|W(t_{k+1})-W(t_k)|\cdot T=0
\end{aligned}
$$

Hence,

$$
\lim_{\|\Pi\|\to 0}\sum_{j=0}^{n-1}(W(t_{j+1})-W(t_j))(t_{j+1}-t_j)=0
$$

which can be written in differential form as $dW(t)dt=0$.

Important results

By the same method, we obtain

$$\lim_{\|\Pi\|\to 0}\sum_{j=0}^{n-1}(t_{j+1}-t_j)^2 \leq \lim_{\|\Pi\|\to 0}\sum_{j=0}^{n-1}\|\Pi\|(t_{j+1}-t_j)$$
$$= \lim_{\|\Pi\|\to 0}\|\Pi\|\cdot T = 0$$

Hence,

$$\lim_{\|\Pi\|\to 0}\sum_{j=0}^{n-1}(t_{j+1}-t_j)^2 = 0$$

which can be written in differential form as $dtdt = 0$.

Important results

Up till now we have obtained three very important results:

$$\boxed{\begin{gathered} dW(t)dW(t) = dt \\ dW(t)dt = 0 \\ dtdt = 0 \end{gathered}}$$

Stochastic calculus

The problem we face when trying to assign meaning to the Itô's integral is that Brownian motion paths cannot be differentiated with respect to time. If $g(t)$ is a differentiable function, then we can define

$$\int_0^T \Delta(t)dg(t) = \int_0^T \Delta(t)g'(t)dt$$

where the RHS is an ordinary (Lebesgue) integral with respect to time. However, this will not work for Brownian motion. We need to try something as follows.

Itô's integral for simple integrands

Definition 2.1.

Let $\Pi = \{t_0, t_1, \cdots, t_n\}$ be a partition of $[0, T]$; i.e., $0 = t_0 < t_1 < \cdots < t_n = T$. A process $\Delta(t)$ is a *simple process* if it is constant in t on each subinterval $[t_j, t_{j+1})$.

Definition 2.2.

Let $W(t)$ be a Brownian motion. The *Itô's integral* $I(t) = \int_0^t \Delta(u)dW(u)$ of the simple process $\Delta(t)$, is defined by, for each $t_j \leq t$

$$I(t) = \sum_{j=0}^{n-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k))$$

Itô's integral for general integrands

We assume that $\Delta(t)$, $t \geq 0$, is adapted to the filtration $\mathcal{F}(t)$, $t \geq 0$. We also assume the square-integrability condition $\mathbb{E}\left(\int_0^T \Delta^2(t)dt\right) < \infty$. (We shall always make such assumptions.) We approximate $\Delta(t)$ by a simple process that equals to $\Delta(t_j)$ at each time t_j and holds this constant over the subinterval $[t_j, t_{j+1})$.

Definition 2.3.

Choose a sequence $\Delta_n(t)$ of simple processes that converges to the continuously varying $\Delta(t)$. Let $W(t)$ be a Brownian motion, and we define the *Itô's integral* by the formula

$$\int_0^t \Delta(t)dW(t) = \lim_{n\to\infty} \int_0^t \Delta_n(u)dW(u)$$

Properties of the Itô's integral

Theorem 2.4.

Let $T > 0$, and let $\Delta(t)$, $0 \le t \le T$, be an adapted stochastic process. Then, $I(t) = \int_0^t \Delta(u)dW(u)$ has the following properties:

(i) **(Continuity)** As a function of the upper limit of integration t, the paths of $I(t)$ are continuous.

(ii) **(Adaptivity)** For each t, $I(t)$ is $\mathcal{F}(t)$-measurable.

(iii) **(Linearity)** If $I(t) = \int_0^t \Delta(u)dW(u)$, $J(t) = \int_0^t \Gamma(u)dW(u)$, then $I(t) \pm J(t) = \int_0^t (\Delta(u) \pm \Gamma(u))dW(u)$; furthermore, for every constant c, $cI(t) = \int_0^t c\Delta(u)dW(u)$.

(iv) **(Martingale)** $I(t)$ is a martingale.

(v) **(Itô isometry)** $\mathbb{E}(I^2(t)) = \mathbb{E}\left(\int_0^t \Delta^2(u)du\right)$.

(vi) **(Quadratic variation)** $[I, I](t) = \int_0^t \Delta^2(u)du$.

Properties of the Itô's integral

Proof.

We prove for simple integrands, and the result follows by taking limits. Properties (i) – (iii) are trivial. We begin with property (iv). Let $0 \leq s \leq t \leq T$ be given. Assume $\exists$ partition points $t_l < t_k$ s.t. $s \in [t_l, t_{l+1})$, $t \in [t_k, t_{k+1})$. Let $D_j = W(t_{j+1}) - W(t_j)$ for $0 \leq j \leq k-1$ and $D_k = W(t) - W(t_k)$. Then we write

$$I(t) = \sum_{j=0}^{l-1} \Delta(t_j) D_j + \Delta(t_l) D_l + \sum_{j=l+1}^{k-1} \Delta(t_j) D_j + \Delta(t_k) D_k$$

Properties of the Itô's integral

Proof (Continued).

- Since $t_l \leq s$, $\forall 0 \leq j \leq l-1$, $\Delta(t_j)D_j$ is $\mathcal{F}(s)$-measurable. Hence, $\mathbb{E}(\Delta(t_j)D_j|\mathcal{F}(s)) = \Delta(t_j)D_j$.
- For the second part,

$$\begin{aligned}
&\mathbb{E}(\Delta(t_l)D_l|\mathcal{F}(s)) \\
&= \Delta(t_l)(\mathbb{E}(W(t_{l+1})|\mathcal{F}(s)) - \mathbb{E}(W(t_l)|\mathcal{F}(s))) \\
&= \Delta(t_l)(W(s) - W(t_l))
\end{aligned}$$

Properties of the Itô's integral

Proof (Continued).

- For $l+1 \leq j \leq k-1$, by the "towering property,"

$$
\begin{aligned}
&\mathbb{E}(\Delta(t_j)D_j|\mathcal{F}(s)) \\
=&\ \mathbb{E}(\mathbb{E}(\Delta(t_j)D_j|\mathcal{F}(t_j))|\mathcal{F}(s)) \\
=&\ \mathbb{E}(\Delta(t_j)(\mathbb{E}(W(t_{j+1})|\mathcal{F}(t_j)) - \mathbb{E}(W(t_j)|\mathcal{F}(t_j)))|\mathcal{F}(s)) \\
=&\ \mathbb{E}(\Delta(t_j)(W(t_j) - W(t_j))|\mathcal{F}(s)) = 0
\end{aligned}
$$

- The fourth part is the same, which equals 0.

Properties of the Itô's integral

Proof (Continued).

Hence, combining the previous four arguments, we conclude that

$$\mathbb{E}(I(t)|\mathcal{F}(s)) = \sum_{j=0}^{l-1} \Delta(t_j)D_j + \Delta(t_l)(W(s) - W(t_l)) = I(s)$$

This proves property (iv). Next we prove property (v). We have

$$I^2(t) = \sum_{j=0}^{k} \Delta^2(t_j)D_j^2 + \sum_{0\le i<j\le k} \Delta(t_i)\Delta(t_j)D_iD_j$$

Properties of the Itô's integral

Proof (Continued).

- Since D_j^2 is independent of $\mathcal{F}(t_j)$ while $\Delta^2(t_j)$ is $\mathcal{F}(t_j)$-measurable, $\mathbb{E}(\Delta^2(t_j)D_j^2) = \mathbb{E}(\Delta^2(t_j))\mathbb{E}(D_j^2)$, thus

$$\mathbb{E}\left(\sum_{j=0}^{k}\Delta^2(t_j)D_j^2\right) = \sum_{j=0}^{k-1}\mathbb{E}(\Delta^2(t_j))(t_{j+1}-t_j) + \mathbb{E}(\Delta^2(t_k))(t-t_k) = \int_0^t \mathbb{E}(\Delta^2(u))du$$

- Since D_j is independent of $\mathcal{F}(t_j)$ while $\Delta(t_i)\Delta(t_j)D_i$ is $\mathcal{F}(t_j)$-measurable,

$$\mathbb{E}(\Delta(t_i)\Delta(t_j)D_iD_j) = \mathbb{E}(\Delta(t_i)\Delta(t_j)D_i)\mathbb{E}(\Delta(t_j)) = 0$$

Properties of the Itô's integral

Proof (Continued).

Hence, combining the previous two arguments, we conclude that

$$\mathbb{E}(I^2(t)) = \int_0^t \mathbb{E}(\Delta^2(u))du + 0 = \mathbb{E}\left(\int_0^t \Delta^2(u)du\right)$$

This proves property (v). Next we prove property (vi). To do this, we first compute the quadratic variation accumulated by the Itô integral on one of the subintervals $[t_j, t_{j+1}]$ on which $\Delta(u)$ is constant $\Delta(t_j)$. We choose $t_j = s_0 < s_1 < \cdots < s_m = t_{j+1}$ as the partition points and consider

$$\sum_{i=0}^{m-1} (I(s_{i+1}) - I(s_i))^2 = \Delta^2(t_j) \sum_{i=0}^{m-1} (W(s_{i+1}) - W(s_i))^2$$

Properties of the Itô's integral

Proof (Continued).

Roughly speaking, since Brownian motion accumulates quadratic variation at rate one per unit time, on each subinterval $[t_j, t_{j+1}]$, the Itô integral accumulates quadratic variation at rate $\Delta^2(t_j)$ per unit time, that is,

$$\lim_{m\to\infty} \sum_{i=0}^{m-1} (I(s_{i+1}) - I(s_i))^2 = \int_{t_j}^{t_{j+1}} \Delta^2(u)du$$

Adding up, we obtain property (v) that

$$[I, I](t) = \int_0^t \Delta^2(u)du$$

□

Itô-Doeblin formula for Brownian motion

The chain rule from ordinary calculus gives us

$$df(W(t)) = f'(W(t))dW(t)$$

However, the correct formula given by the Itô-Doeblin formula is

$$df(W(t)) = f'(W(t))dW(t) + \frac{1}{2}f''(W(t))dt$$

Why is there an extra term?

Itô-Doeblin formula for Brownian motion

Theorem 2.5 (Itô-Doeblin formula for Brownian motion).

Let $f(t,x)$ be a function for which the partial derivatives $f_t(t,x)$, $f_x(t,x)$, and $f_{xx}(t,x)$ are defined and continuous, and let $W(t)$ be a Brownian motion. Then, for every $T \geq 0$,

$$f(T, W(T)) = f(0, W(0)) + \int_0^T f_t(t, W(t))dt + \int_0^T f_x(t, W(t))dW(t) + \frac{1}{2}\int_0^T f_{xx}(t, W(t))dt$$

Itô-Doeblin formula for Brownian motion

Proof (Sketch).

Fix $T > 0$, and let $\Pi = \{t_0, t_1, \cdots, t_n\}$ be a partition of $[0, T]$. Then, bring $\|\Pi\| \to 0$ and apply Taylor's formula, we obtain

$$f(T, W(T)) - f(0, W(0)) = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} \Sigma_j$$

where, if we omit the arguments

$$\begin{aligned}\Sigma_j &= f_t(t_j, W(t_j))(t_{j+1} - t_j) + f_x(t_j, W(t_j))(W(t_{j+1}) - W(t_j)) \\ &+ \frac{1}{2} f_{tt}(t_j, W(t_j))(t_{j+1} - t_j)^2 + \frac{1}{2} f_{xx}(t_j, W(t_j))(W(t_{j+1}) - W(t_j))^2 \\ &+ f_{tx}(t_j, W(t_j))(t_{j+1} - t_j)(W(t_{j+1}) - W(t_j)) + \text{higher order terms}\end{aligned}$$

Itô-Doeblin formula for Brownian motion

Proof (Sketch, Continued).

For the sake of simplicity, we write $f(t, W(t))$ as f and so on. If we write in differential form, then we will have

$$df = f_t dt + f_x dW(t) + \frac{1}{2} f_{tt} dtdt + \frac{1}{2} f_{xx} dW(t) dW(t) + f_{tx} dtdW(t)$$

Recall that $dW(t)dW(t) = dt$, $dtdt = 0$, and $dtdW(t) = 0$. This is the reason why we omit the higher order terms. Also, the answer will simplify to $df = f_t dt + f_x dW(t) + \frac{1}{2} f_{xx} dt$, which is exactly our desired result in differential form.

□

Itô-Doeblin formula for an Itô process

Definition 2.6.

Let $W(t)$, $t \geq 0$, be a Brownian motion, and let $\mathcal{F}(t)$, $t \geq 0$, be an associated filtration. An *Itô process* is a stochastic process of the form

$$X(t) = X(0) + \int_0^t \Delta(u)dW(u) + \int_0^t \Theta(u)du$$

where $X(0)$ is nonrandom and $\Delta(u)$ and $\Theta(u)$ are adapted stochastics processes.

We assume that $\mathbb{E}\left(\int_0^t \Delta^2(u)du\right) < \infty$ and $\int_0^t |\Theta(u)|du < \infty$ so that the integrals on the RHS are defined and the Itô integral is a martingale. We shall always make such integrability assumptions.

Itô-Doeblin formula for an Itô process

Lemma 2.7.

Let X be an Itô process. Then $[X, X](t) = \int_0^t \Delta^2(u)du$.

Proof (Sketch).

We write in differential form as $dX(t) = \Delta(t)dW(t) + \Theta(t)dt$. Then, $dX(t)dX(t) = \Delta^2(t)dW(t)dW(t) + 2\Delta(t)\Theta(t)dW(t)dt + \Theta^2(t)dtdt = \Delta^2(t)dt$. (This says that, at each time t, the Itô process X accumulates quadratic variation at rate $\Delta^2(t)$ per unit time.) This is exactly our desired result in differential form.

□

Itô-Doeblin formula for an Itô process

Definition 2.8.

Let $X(t)$, $t \geq 0$, be an Itô process as described in Definition 2.6, and let $\Gamma(t)$, $t \geq 0$, be an adapted process. We define the integral with respect to an Itô process as

$$\int_0^t \Gamma(u)dX(u) = \int_0^t \Gamma(u)\Delta(u)dW(u) + \int_0^t \Gamma(u)\Theta(u)du$$

Itô-Doeblin formula for an Itô process

Theorem 2.9 (Itô-Doeblin formula for an Itô process).

Let $X(t)$, $t \geq 0$, be an Itô process as described in Definition 2.6, and let $f(t,x)$ be a function for which the partial derivatives $f_t(t,x)$, $f_x(t,x)$, and $f_{xx}(t,x)$ are defined and continuous. Then, for every $T \geq 0$,

$$\begin{aligned} f(T, X(T)) &= f(0, X(0)) \\ &\quad + \int_0^T f_t(t, X(t))dt + \int_0^T f_x(t, X(t))\Delta(t)dW(t) \\ &\quad + \int_0^T f_X(t, X(t))\Theta(t)dt + \frac{1}{2}\int_0^T f_{xx}(t, X(t))\Delta^2(t)dt \end{aligned}$$

Itô-Doeblin formula for an Itô process

Proof (Sketch, Continued).

For the sake of simplicity, we write $f(t, X(t))$ as f and so on. If we write in differential form, by Taylor's formula and the same argument as in Theorem 2.5 to omit the higher order terms,

$$df = f_t dt + f_x dX(t) + \frac{1}{2} f_{tt} dtdt + \frac{1}{2} f_{xx} dX(t) dX(t) + f_{tx} dtdX(t)$$

Note that
- $dtdt = 0$;
- $dX(t) = \Delta(t) dW(t) + \Theta(t) dt$;
- $dX(t) dX(t) = d[X, X](t) = \Delta^2(t) dt$;
- $dtdX(t) = \Delta(t) dtdW(t) + \Theta(t) dtdt = 0$.

Hence, $df = f_t dt + f_x \Delta(t) dW(t) + f_x \Theta(t) dt + \frac{1}{2} f_{xx} \Delta^2(t) dt$, which is exactly our desired result in differential form.

□

Evolution of portfolio value

Consider an agent who at each time t has a portfolio valued at $X(t)$. This portfolio invests in:

- a money account paying a constant rate of interest r; and
- a stock modeled by the geometric Brownian motion $S(t)$ s.t. $dS(t) = \alpha S(t)dt + \sigma S(t)dW(t)$.

Evolution of portfolio value

Suppose at each time t the investor holds $\Delta(t)$ shares of stock, which is random but adapted to the filtration associated with the Brownian motion $W(t)$, $t \geq 0$. We can compute the following:

- the capital gain on the stock position: $\Delta(t)dS(t)$; and
- the interest earnings on the cash position: $r(X(t) - \Delta(t)dS(t))dt$.

Hence, the portfolio value

$$\begin{aligned} dX(t) &= \Delta(t)dS(t) + r(X(t) - \Delta(t)dS(t))dt \\ &= \Delta(t)(\alpha S(t)dt + \sigma S(t)dW(t)) + r(X(t) - \Delta(t)dS(t))dt \\ &= rX(t)dt + \Delta(t)(\alpha - r)S(t)dt + \Delta(t)\sigma S(t)dW(t) \end{aligned}$$

Evolution of portfolio value

To determine the present value of a stream of payments that is to be received in the future, we shall often consider the discounted stock price $e^{-rt}S(t)$ and the discounted portfolio value $e^{-rt}X(t)$. Apply the Itô-Doeblin formula with $f(t,x) = e^{-rt}x$, we obtain

$$\begin{aligned}
d(e^{-rt}S(t)) &= -re^{-rt}S(t)dt + e^{-rt}dS(t) + 0 \\
&= (\alpha - r)e^{-rt}S(t)dt + \sigma e^{-rt}S(t)dW(t) \\
d(e^{-rt}X(t)) &= -re^{-rt}X(t)dt + e^{-rt}dX(t) + 0 \\
&= \Delta(t)(\alpha - r)e^{-rt}S(t)dt + \Delta(t)\sigma e^{-rt}S(t)dW(t) \\
&= \Delta(t)d(e^{-rt}S(t))
\end{aligned}$$

Evolution of option value

Consider a European call option that pays $(S(T)-K)^+$ at time T. The strike price K is some nonnegative constant. Let $c(t,x)$ denote the value of the call at time t if the stock price at that time is $S(t)=x$. Our goal is to find a profitable $c(t,x)$. We compute its derivative by the Itô-Doeblin formula. For the sake of simplicity, we write $c(t,S(t))$ as c and so on. We shall obtain

$$\begin{aligned} dc &= c_t dt + c_x dS(t) + \tfrac{1}{2} c_{xx} dS(t) dS(t) \\ &= c_t dt + \alpha S(t) c_x dt + \sigma S(t) c_x dW(t) + \tfrac{1}{2}\sigma^2 S^2(t) c_{xx} dt \end{aligned}$$

Notice that the last summand follows from computing $dS(t)dS(t)$
$= \alpha^2 S^2(t) dtdt + \alpha\sigma S^2(t) dtdW(t) + \sigma^2 S^2(t) dW(t) dW(t)$
$= \sigma^2 S^2(t) dt.$

Evolution of option value

Same as before, we shall often consider the discounted option price $e^{-rt}c(t, S(t))$. Let $f(t, x) = e^{-rt}x$. By the Itô-Doeblin formula,

$$\begin{aligned} d(e^{-rt}c) &= -re^{-rt}cdt + e^{-rt}dc + 0 \\ &= -re^{-rt}cdt + e^{-rt}c_t dt + \alpha e^{-rt}S(t)c_x dt \\ &\quad + \sigma e^{-rt}S(t)c_x dW(t) + \tfrac{1}{2}\sigma^2 e^{-rt}S^2(t)c_{xx}dt \\ &= e^{-rt}(-rc + c_t + \alpha S(t)c_x + \tfrac{1}{2}\sigma^2 S^2(t)c_{xx})dt \\ &\quad + e^{-rt}\sigma S(t)c_x dW(t) \end{aligned}$$

Black-Scholes-Merton equation

A (short option) hedging portfolio starts with some initial capital $X(0)$ and invests in the stock and money market account so that the portfolio value $X(t)$ at each time $t \in [0, T]$ agrees with the value of call $c(t, S(t))$. Hence, $d(e^{-rt}X(t)) = d(e^{-rt}c(t, S(t)))$ suffices since the initial values $X(0) = c(0, S(0))$. Thus we require

$$\begin{aligned}&\Delta(t)(\alpha - r)S(t)dt + \Delta(t)\sigma S(t)dW(t)\\&\quad = (-rc + c_t + \alpha S(t)c_x + \tfrac{1}{2}\sigma^2 S^2(t)c_{xx})dt + \sigma S(t)c_x dW(t)\end{aligned}$$

Black-Scholes-Merton equation

We need to equate the dt terms and $dW(t)$ terms respectively.

$$\begin{cases} \Delta(t)\sigma S(t) = \sigma S(t) c_x \implies \Delta(t) = c_x \\ \Delta(t)(\alpha - r)S(t) = -rc + c_t + \alpha S(t) c_x + \frac{1}{2}\sigma^2 S^2(t) c_{xx} \end{cases}$$

Substituting $\Delta(t)$ in the second equation by c_x according to the first equation, we obtain the **Black-Scholes-Merton partial differential equation**:

$$\boxed{c_t(t,x) + rxc_x(t,x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t,x) = rc(t,x), \quad t \in [0, T)}$$

with boundary condition $c(T,x) = (x-K)^+$, and $x \geq 0$.

Black-Scholes-Merton equation

The corresponding solution to Black-Scholes-Merton equation is:

$$\boxed{c(t,x) = xN(d_+(T-t,x)) - Ke^{-r(T-t)}N(d_-(T-t,x))}$$

where $0 \leq t \leq T$, $x > 0$, N is the cumulative standard normal distribution, and

$$\boxed{d_\pm(\tau,x) = \frac{1}{\sigma\sqrt{\tau}}\left(\log\frac{x}{K} + \left(r \pm \frac{\sigma^2}{2}\right)r\right)}$$

We shall sometimes use the notation of the Black-Scholes-Merton function $\mathrm{BSM}(\tau, x; K, r, \sigma)$:

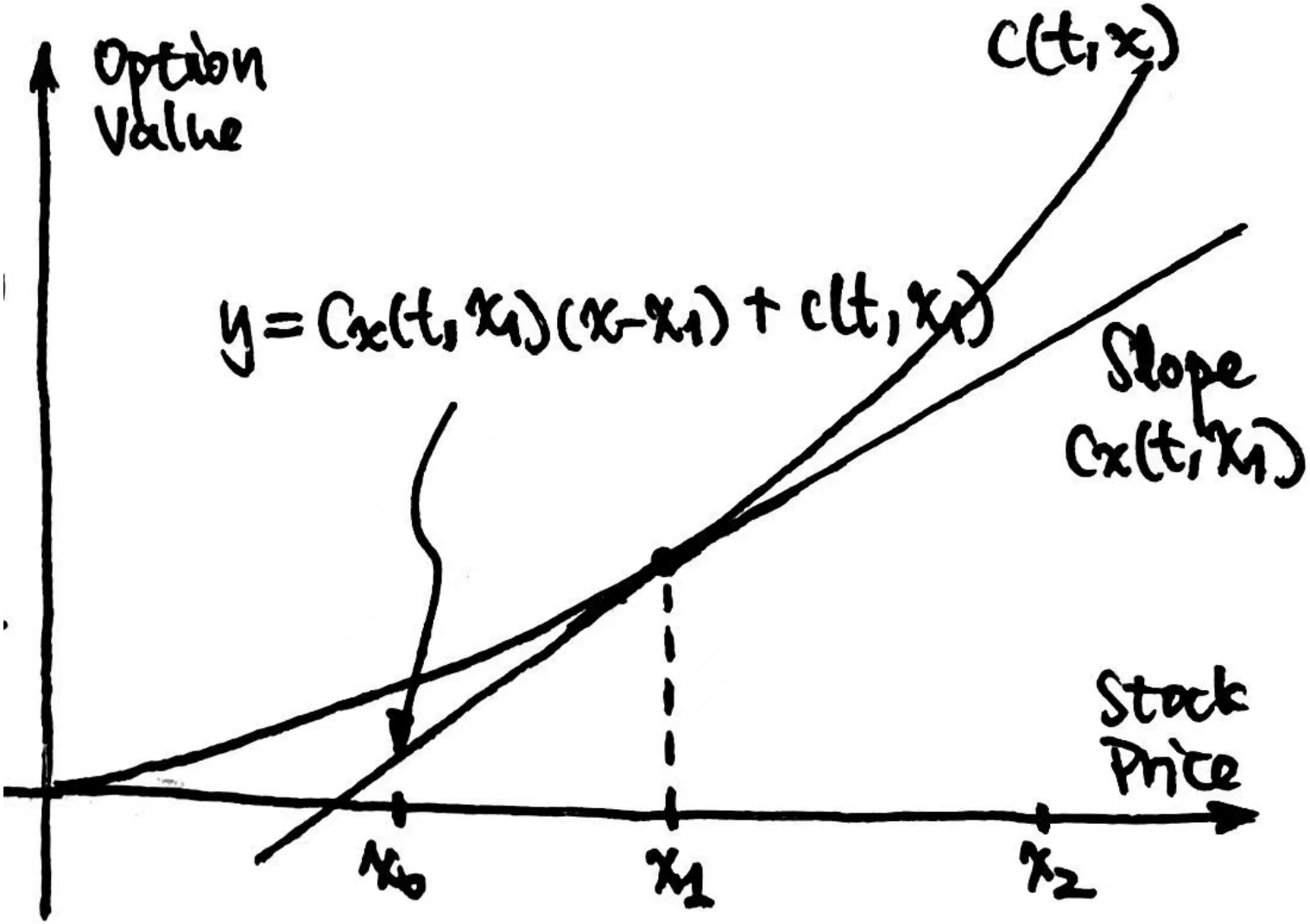
$$\mathrm{BSM}(\tau, x; K, r, \sigma) = xN(d_+(\tau,x)) - Ke^{-r\tau}N(d_-(\tau,x))$$

Black-Scholes-Merton equation

The partial derivatives of the function $c(t,x)$ are called the Greeks. Substitute $\tau = T - t$, then

- $c_x(t,x) = N(d_+(\tau,x))$ is called *delta*, which is positive;
- $c_t(t,x) = -rKe^{-r\tau}N(d_-(\tau,x)) - \frac{\sigma x}{2\sqrt{\tau}}N'(d_+(\tau,x))$ is called *theta*, which is negative; and
- $c_{xx}(t,x) = N'(d_+(\tau,x))\frac{\partial}{\partial x}d_+(\tau,x) = \frac{1}{\sigma x\sqrt{\tau}}N'(d_+(T-t,x))$ is called *gamma*, which is positive.

Black-Scholes-Merton equation



Intuition for an alternative solution

We want to obtain $V(t)$, $0 \leq t \leq T$ directly from $\mathbb{E}(V(T)|\mathcal{F}(t))$ s.t. $V(t) = \mathbb{E}(V(T)|\mathcal{F}(s))$ for all $0 \leq t \leq T$. However, $V(T)$ is not a martingale in real-life probability, so we try to find another probability measure $\tilde{\mathbb{P}}$ s.t. $V(T)$ is a martingale under $\tilde{\mathbb{P}}$ and $V(t) = \mathbb{E}^{\tilde{\mathbb{P}}}(V(T)|\mathcal{F}(s))$.

Radon-Nikodým derivative

Definition 3.1.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\tilde{\mathbb{P}}$ be another probability measure on $(\Omega, \mathcal{F})$ that is equivalent to $\mathbb{P}$, and let Z be an almost surely positive random variable with $\mathbb{E}(Z) = 1$ that relates $\mathbb{P}$ and $\tilde{\mathbb{P}}$ via

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega) \quad \text{for all } A \in \mathcal{F}$$

Then Z is called the *Radon-Nikodým derivative* of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$, and we write

$$Z = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$$

Radon-Nikodým derivative process

We can define the Radon-Nikodým derivative process

$$Z(t) = \mathbb{E}(Z|\mathcal{F}(t)), \quad 0 \leq t \leq T$$

The Radon-Nikodým derivative process is a martingale because $\mathbb{E}(Z(t)|\mathcal{F}(S)) = \mathbb{E}(\mathbb{E}(Z|\mathcal{F}(t))|\mathcal{F}(s)) = \mathbb{E}(Z|\mathcal{F}(s)) = Z(s)$.

Lemma 3.2.

Let s and t satisfying $0 \leq s \leq t \leq T$ be given and let Y be an $\mathcal{F}(t)$-measurable random variable. Then

$$\tilde{\mathbb{E}}(Y|\mathcal{F}(s)) = \frac{1}{Z(s)}\mathbb{E}(YZ(t)|\mathcal{F}(s))$$

Girsanov's Theorem

Theorem 3.3 (Girsanov, one dimension).

Let $W(t)$, $0 \le t \le T$, be a Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let $\mathcal{F}(t)$, $0 \le t \le T$, be a filtration for this Brownian motion. Let $\Theta(t)$, $0 \le t \le T$, be an adapted process. Define

$$Z(t) = \exp\left(-\int_0^t \Theta(u)dW(u) - \frac{1}{2}\int_0^t \Theta^2(u)du\right)$$
$$\tilde{W}(t) = W(t) + \int_0^t \Theta(u)du$$

and assume that $\mathbb{E}\left(\int_0^T \Theta^2(u)Z^2(u)du\right) < \infty$. Set $Z = Z(T)$. Then $\mathbb{E}(Z) = 1$ and under the probability measure $\tilde{\mathbb{P}}$ related to $\mathbb{P}$ by Radon-Nikodým derivative Z, the process $\tilde{W}(t)$, $0 \le t \le T$, is a Brownian motion.

Girsanov's Theorem

Theorem 3.4 (Lévy, one dimension).

Let $M(t)$, $t \geq 0$, be a martingale relative to a filtration $\mathcal{F}(t)$, $t \geq 0$. Assume that $M(0) = 0$, $M(t)$ has continuous paths, and $[M, M](t) = t$ for all $r \geq 0$. Then $M(t)$ is a Brownian motion.

Proof (of Girsanov, one dimension).

We want to meet the conditions required by Lévy's Theorem. The second and third conditions are easy to follow:

- $\tilde{W}(0) = W(0) + \int_0^0 \Theta(u)du = 0$; and
- $d\tilde{W}(t)d\tilde{W}(t) = (dW(t) + \Theta(t)dt)^2 = dW(t)dW(t)$ $+2\Theta(t)dW(t)dt + \Theta^2(t)dtdt = dt$.

Girsanov's Theorem

Proof (Continued).

It remains to show that $\tilde{W}(t)$ is a martingale under $\tilde{\mathbb{P}}$. We first observe that $Z(t)$ is a martingale under $\mathbb{P}$. With

$$X(t) = -\int_0^t \Theta(u)dW(u) - \frac{1}{2}\int_0^t \Theta^2(u)du$$

and $f(x) = e^x$ so that $f'(x) = e^x$ and $f''(x) = e^x$, we have

$$\begin{aligned} dZ(t) = df(X(t)) &= f'(X(t))dX(t) + \tfrac{1}{2}f''(X(t)dX(t)dX(t) \\ &= e^{X(t)}(-\Theta(t)dW(t) - \tfrac{1}{2}\Theta^2(t)dt) + \tfrac{1}{2}e^{X(t)}\Theta^2(t)dt \\ &= -\Theta(t)Z(t)dW(t) \end{aligned}$$

Girsanov's Theorem

Proof (Continued).

Integrating both sides of the previous equation, we see that

$$Z(t) = Z(0) - \int_0^t \Theta(u) Z(u) dW(u)$$

Because Itô integrals are martingales, $Z(t)$ is a martingale. In particular, $\mathbb{E}(Z) = \mathbb{E}(Z(T)) = Z(0) = 1$. Then we will obtain $Z(t) = \mathbb{E}(Z(T)|\mathcal{F}(t)) = \mathbb{E}(Z|\mathcal{F}(t))$, $0 \leq t \leq T$.
We next show that $\tilde{W}(t)Z(t)$ is a martingale under $\mathbb{P}$. To see this, we need to state Itô's product rule.

Girsanov's Theorem

Corollary 3.5 (Itô's product rule).

Let $X(t)$ and $Y(t)$ be Itô processes. Then

$$d(X(t)Y(t)) = X(t)dY(t) + Y(t)dX(t) + dX(t)dY(t)$$

Proof (Continued).

Apply Itô's product rule, we obtain

$$\begin{aligned} d(\tilde{W}(t)Z(t)) &= \tilde{W}(t)dZ(t) + Z(t)d\tilde{W}(t) + d\tilde{W}(t)dZ(t) \\ &= -\tilde{W}(t)\Theta(t)Z(t)dW(t) + Z(t)dW(t) + Z(t)\Theta(t)dt \\ &\quad + (dW(t) + \Theta(t)dt)(-\Theta(t)Z(t)dW(t)) \\ &= (-\tilde{W}(t)\Theta(t) + 1)Z(t)dW(t) \end{aligned}$$

Girsanov's Theorem

Proof (Continued).

Since the previous expression has no dt term, the process $\tilde{W}(t)Z(t)$ is indeed a martingale under $\mathbb{P}$.
Now let $0 \leq s \leq t \leq T$ be given. Lemma 3.2 and the martingale property for $\tilde{W}(t)Z(t)$ under $\mathbb{P}$ imply

$$\begin{aligned}\tilde{\mathbb{E}}(\tilde{W}(t)|\mathcal{F}(s)) &= \frac{1}{Z(s)}\mathbb{E}(\tilde{W}(t)Z(t)|\mathcal{F}(s)) \\ &= \frac{1}{Z(s)}Z(s)\mathbb{E}(\tilde{W}(t)|\mathcal{F}(s)) = \tilde{W}(s)\end{aligned}$$

This shows that $\tilde{W}(t)$ is a martingale under $\tilde{\mathbb{P}}$. This meets the first condition required by Lévy's Theorem, so the proof is complete. □

Stock under the risk-neutral measure

Let $W(t)$, $0 \leq t \leq T$, be a Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let $\mathcal{F}(t)$, $0 \leq t \leq T$, be a filtration for this Brownian motion. Here T is a fixed final time. Consider a stock price process s.t. $dS(t) = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t)$. We have $S(t)$ as the generalized Brownian motion

$$S(t) = S(0)\exp\left(\int_0^t \sigma(s)dW(s) + \int_0^t \left(\alpha(s) - \frac{1}{2}\sigma^2(s)\right)ds\right)$$

Stock under the risk-neutral measure

The discounted stock process $D(t)S(t)$ is then expressed as

$$S(0)\exp\left(\int_0^t \sigma(s)dW(s) + \int_0^t \left(\alpha(s) - R(s) - \frac{1}{2}\sigma^2(s)\right) ds\right)$$

and its differential is

$$\begin{aligned} d(D(t)S(t)) &= (\alpha(t) - R(t))D(t)S(t)dt + \sigma(t)D(t)S(t)dW(t) \\ &= \sigma(t)D(t)S(t)(\Theta(t)dt + dW(t)) \end{aligned}$$

where $D(t)$ is the discounted process, $\alpha(t)$ is the mean rate of return, $\sigma(t)$ is the volatility, R is the interest rate, and

$$\Theta(t) = \frac{\alpha(t) - R(t)}{\sigma(t)}$$

Stock under the risk-neutral measure

Let $dW(t) = -\Theta(t)dt + d\tilde{W}(t)$, so by previous computations, $d(D(t)S(t)) = \sigma(t)D(t)S(t)d\tilde{W}(t)$. By Girsanov's Theorem (Theorem 3.3), we know that $\tilde{W}(t)$ is a Brownian motion under measure $\tilde{\mathbb{P}}$. Also, by Theorem 2.4 (iv), $D(t)S(t)$ is a martingale under measure $\tilde{\mathbb{P}}$. By replacing $dW(t)$, we now express $dS(t)$ as $dS(t) = R(t)S(t)dt + \sigma(t)S(t)d\tilde{W}(t)$. Thus,

$$S(t) = S(0)\exp\left(\int_0^t \sigma(s)d\tilde{W}(s) + \int_0^t \left(R(s) - \frac{1}{2}\sigma^2(s)\right)ds\right)$$

Deriving the Black-Scholes-Merton formula

With constant σ and r, the previous formula becomes

$$S(t) = S(0)\exp\left(\sigma\tilde{W}(t) + \left(r - \frac{1}{2}\sigma^2\right)t\right)$$

and we may thus write

$$\begin{aligned} S(T) &= S(t)\exp\left(\sigma(\tilde{W}(T) - \tilde{W}(t)) + \left(r - \frac{1}{2}\sigma^2\right)\tau\right) \\ &= S(t)\exp\left(-\sigma\sqrt{\tau}Y + \left(r - \frac{1}{2}\sigma^2\right)\tau\right) \end{aligned}$$

where Y is the standard normal random variable

$$Y = -\frac{\tilde{W}(T) - \tilde{W}(t)}{\sqrt{T-t}}, \quad \tau \text{ is the "time to expiration" } T - t.$$

Deriving the Black-Scholes-Merton formula

We have the equation $D(t)V(t) = \tilde{\mathbb{E}}(D(T)V(T)|\mathcal{F}(t))$ by the martingale property of $D(t)V(t)$. Now, set the discounted process $D(t) = e^{-rt}$ and $V(T) = (S(T) - K)^+$, which is the European call option. Also, we denote $c(t, x) = V(t)$ where $S(0) = x$. Then we get:

$$c(t,x) = \tilde{\mathbb{E}}\left(e^{-r\tau}\left(x\exp\left(-\sigma\sqrt{\tau}Y + \left(r - \frac{1}{2}\sigma^2\right)\tau\right) - K\right)^+\right)$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-r\tau}\left(x\exp\left(-\sigma\sqrt{\tau}Y + \left(r - \tfrac{1}{2}\sigma^2\right)\tau\right) - K\right)^+ e^{-\frac{1}{2}y^2}dy$$

Deriving the Black-Scholes-Merton formula

The integrand

$$\left(x \exp\left(-\sigma\sqrt{\tau}Y + \left(r - \frac{1}{2}\sigma^2\right)\tau\right) - K\right)^+$$

is positive iff

$$y < d_-(\tau, x) = \frac{1}{\sigma\sqrt{\tau}}\left(\log\frac{x}{K} + \left(r - \frac{1}{2}\sigma^2\right)\tau\right)$$

Now it remains to explicitly compute the formula of $c(t, x)$.

Deriving the Black-Scholes-Merton formula

Therefore, $c(t,x)$

$$
\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau,x)} x \exp\left(-\frac{y^2}{2} - \sigma\sqrt{\tau}Y - \frac{\sigma^2\tau}{2}\right) dy \\
&\quad - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau,x)} K \exp\left(-\frac{y^2}{2} - r\tau\right) dy \\
&= \frac{x}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau,x)} \exp\left(-\frac{1}{2}(y+\sigma\sqrt{\tau})^2\right) dy - e^{-r\tau} K N(d_-(\tau,x)) \\
&= \frac{x}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau,x)+\sigma\sqrt{\tau}} \exp\left(-\frac{z^2}{2}\right) dz - e^{-r\tau} K N(d_-(\tau,x))
\end{aligned}
$$

(Continued)

Deriving the Black-Scholes-Merton formula

(Continued)

Following the previous computation, we finally obtain

$$\boxed{c(t,x) = xN(d_+(\tau,x)) - e^{-r\tau}KN(d_-(\tau,x))}$$

where

$$\boxed{d_+(\tau,x) = d_-(\tau,x) + \sigma\sqrt{\tau} = \frac{1}{\sigma\sqrt{tau}}\left(\log\frac{x}{K} + \left(r + \frac{1}{2}\sigma^2\right)\tau\right)}$$